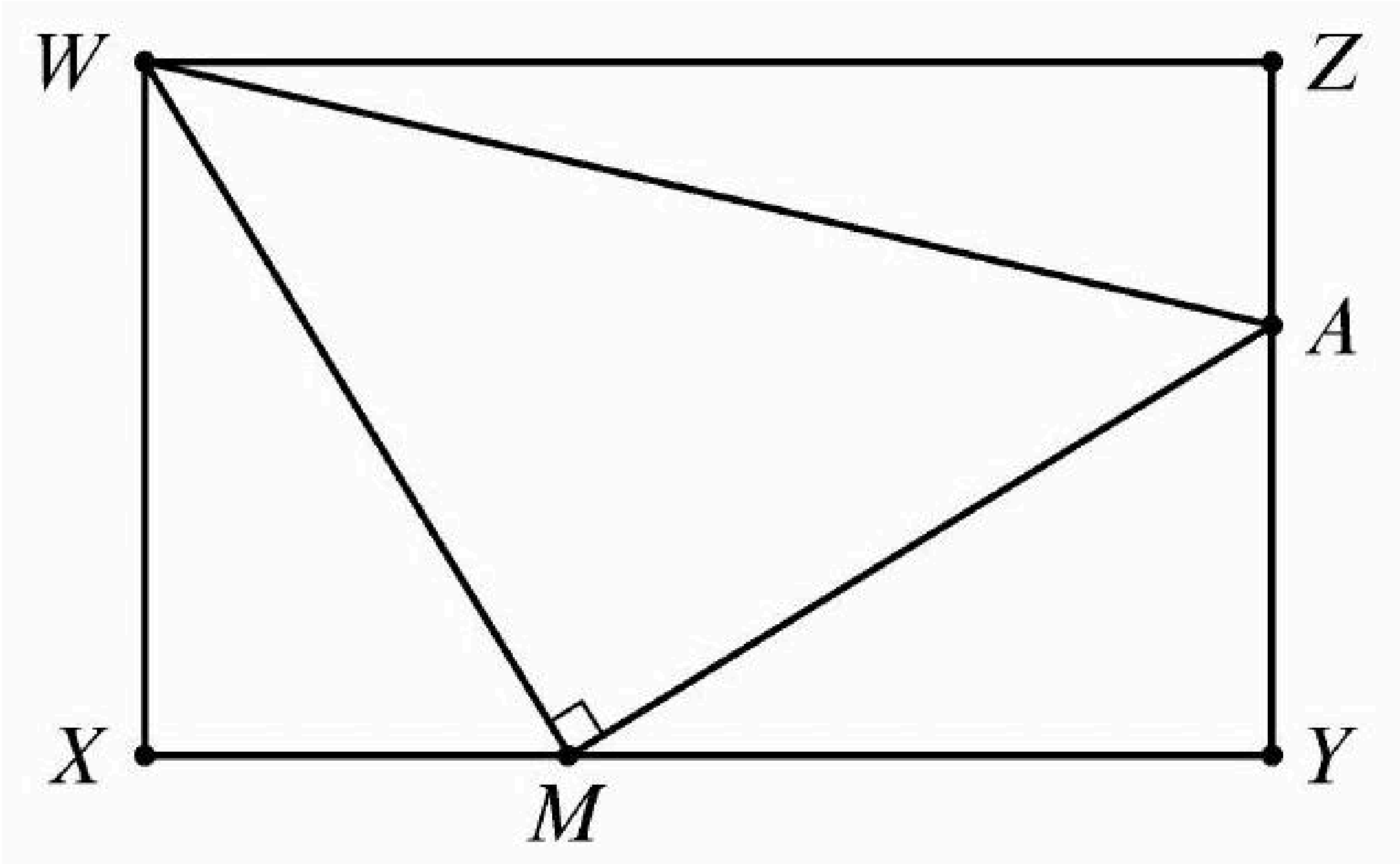
2024 AMC 10B Problem 11 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10B Problem 11. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10B Problem 11

Problem:

In the figure below $WXYZ$ is a rectangle with $WX = 4$ and $WZ = 8$. Point M lies on $\overline{XY}$, point A lies on $\overline{YZ}$, and $\angle WMA$ is a right angle. The areas of triangles $\triangle WXM$ and $\triangle WAZ$ are equal. What is the area of $\triangle WMA$?



Answer Choices:

A. 13

B. 14

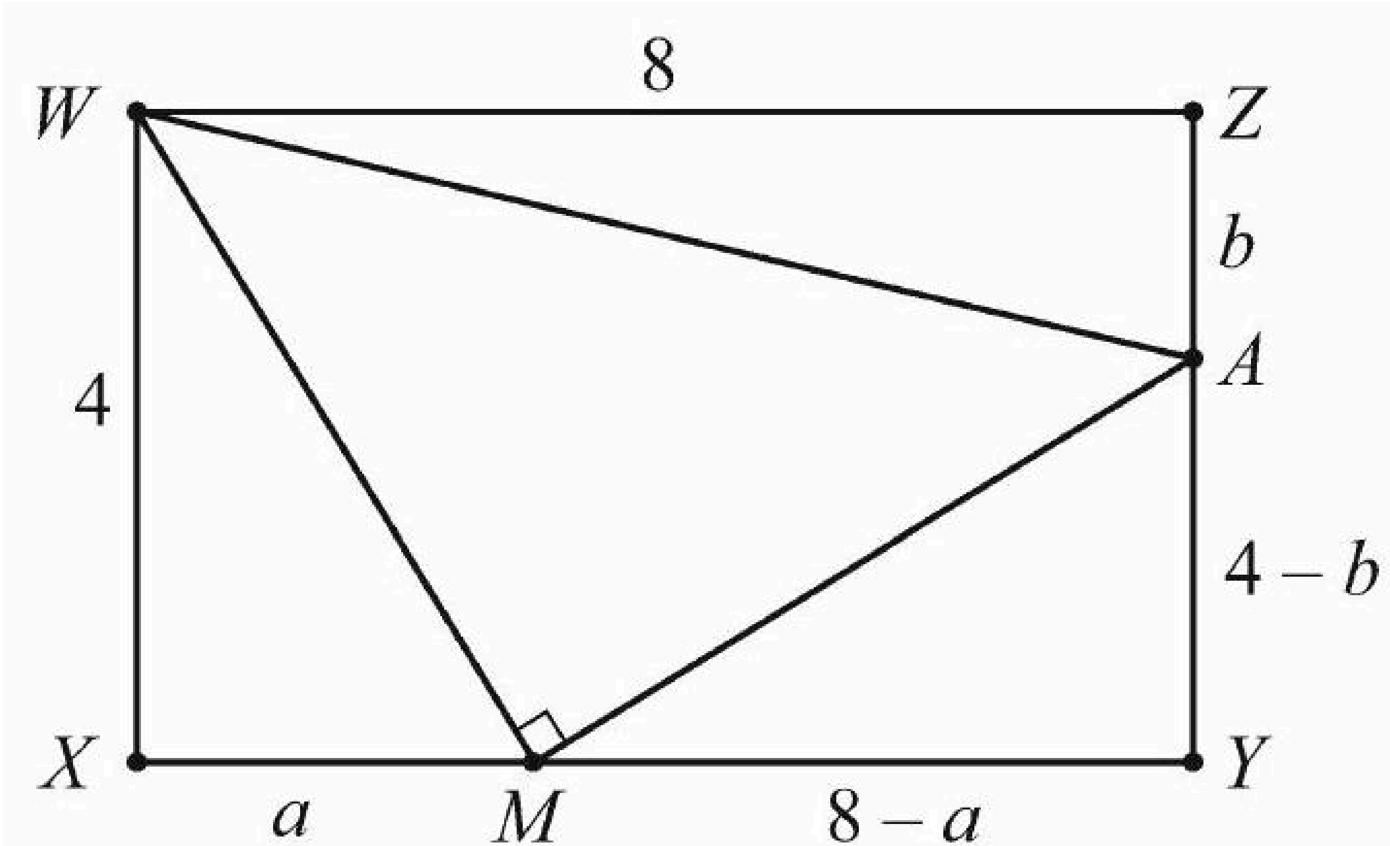
C. 15

D. 16

E. 17

Solution:

Label the diagram as shown, where $MX = a$ and $ZA = b$.



The Pythagorean Theorem on $\triangle WMA$ gives $WM^2 + MA^2 = WA^2$, which implies that

$$4^2 + a^2 + (8-a)^2 + (4-b)^2 = 8^2 + b^2$$

Expanding and simplifying yields $a^2 - 8a - 4b + 16 = 0$. Because the areas of triangles $\triangle WXM$ and $\triangle WAZ$ are equal, $\frac{1}{2} \cdot 4a = \frac{1}{2} \cdot 8b$, so $a = 2b$. Substituting into the previous equation and factoring\
gives $4(b-1)(b-4) = 0$. Therefore $b = 1$ or $b = 4$. But $b = 4$ would require $A = Y = M$, and $\angle WMA$ would not exist, so it must be that $b = 1$ and $a = 2$. The area of $\triangle WMA$ can be found by subtracting the three other triangle areas from the area of the rectangle:

$$8 \cdot 4 - \frac{1}{2} \cdot 4 \cdot 2 - \frac{1}{2} \cdot 6 \cdot 3 - \frac{1}{2} \cdot 8 \cdot 1 = 32 - 4 - 9 - 4 = \text{(C)}\boxed{15}$$